

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of  
Computer Science and Engineering**

**ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS**

|                      |                                                                                                                                                            |                     |                                               |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------|
| <b>Cour</b>          | CSA06                                                                                                                                                      | <b>Cou</b>          | Design and Analysis of Algorithm              |
| <b>C</b>             | CO1 – Precision (BL1, BL2, BL3)                                                                                                                            | <b>Assessment:</b>  | Assessment Tool 1 – ApplicationBase Questions |
| <b>Weightage:</b>    | 40% of C                                                                                                                                                   | <b>Total Marks:</b> | 100 (2 Questions × 50 M                       |
| <b>CO1 State</b>     | <i>Determine algorithm efficiency using asymptotic notations and mathematical techniques includingMas Theorem and substitution method. (BL1, BL2, BL3)</i> |                     |                                               |
| <b>Student Name:</b> | L ARAVIND                                                                                                                                                  | <b>Reg. No.:</b>    | 192424080                                     |
| <b>Section</b>       | / Batch: A / 2024                                                                                                                                          | <b>Date:</b>        | 17/07/2026                                    |

**Instructions to Students**

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks. • Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

|                                    |                                                            |    |                         |
|------------------------------------|------------------------------------------------------------|----|-------------------------|
|                                    |                                                            |    |                         |
| <b>Application-Based Questions</b> | Apply Master Theorem and asymptotican real-world scenarios | Q1 | –<br>Q1–50<br>Q2<br>–50 |
|                                    |                                                            |    |                         |

**SECTION A – APPLICATION BASED QUESTIONS**

**Q1. Application: Weather Forecasting Simulation System**

A weather simulation system uses two algorithms with complexities  $O(n \log n)$  and  $O(n^2 \log n)$ . Analyze both algorithms using asymptotic notation and compare their growth rates. Explain how each algorithm performs as the dataset size increases. Determine which algorithm is more efficient for large-scale simulations. Justify your answer based on computational efficiency and scalability.

```

main.py
1 n = int(input("Enter dataset size: "))
2
3 # Approximate log2(n)
4 log = 0
5 temp = n
6 while temp > 1:
7     temp //= 2
8     log += 1
9
10 algo1 = n * log      # O(n log n)
11 algo2 = n * n * log  # O(n^2 log n)
12
13 print("Algorithm 1 (O(n log n)) =", algo1)
14 print("Algorithm 2 (O(n^2 log n)) =", algo2)
15
16 if algo1 < algo2:
17     print("Algorithm 1 is more efficient for large datasets.")
18 else:
19     print("Algorithm 2 is more efficient.")

```

```

Output
Enter dataset size: 15
Algorithm 1 (O(n log n)) = 45
Algorithm 2 (O(n^2 log n)) = 675
Algorithm 1 is more efficient for large datasets

=== Code Execution Successful ===

```

**Q2. Application: Online Learning Recommendation System** An online learning platform processes user interaction data using the recurrence  $T(n) = T(n-1) + n$ . Apply the substitution method to expand the recurrence step-by-step and derive the final time complexity. Clearly show intermediate steps and simplify the expression. Analyze how the algorithm performs when the number of users grows rapidly. Interpret the efficiency using asymptotic notation and discuss whether the system is scalable for large educational platforms.

```

main.py
1 n = int(input("Enter number of users (n): "))
2
3 print("\nExpansion:")
4 print("T(n) = T(n-1) + n")
5 print("    = T(n-2) + (n-1) + n")
6 print("    = T(n-3) + (n-2) + (n-1) + n")
7 print("    = ...")
8 print("    = T(1) + 2 + 3 + ... + n")
9
10 total = n * (n + 1) // 2
11
12 print("\nSum =", total)
13 print("Final Time Complexity = O(n^2)")

```

```

Output
Enter number of users (n): 10

Expansion:
T(n) = T(n-1) + n
      = T(n-2) + (n-1) + n
      = T(n-3) + (n-2) + (n-1) + n
      = ...
      = T(1) + 2 + 3 + ... + n

Sum = 55
Final Time Complexity = O(n^2)

=== Code Execution Successful ===

```

### Code of Conduct

I L ARAVIND [192424080] (Name / Reg No) certify that this submission is my original work and that I have adhered guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

to the

### RUBRICS FOR CALCULATION

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|  |  |  |  |  |  |
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|                                  |     |                                                                   |                                                                              |                                                   |                                                        |
|----------------------------------|-----|-------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20% | Clearly understands problem and identifies all requirements       | Good understanding of the problem and accurately understands with minor gaps | Basic understanding; some requirements unclear    | Poor understanding or misinterpretation of the problem |
| Application Concepts             | 30% | Accurately applies appropriate concepts to solve complex problems | Applies relevant concepts correctly with minor errors                        | Applies basic concepts with very limited accuracy | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20% | Logical, well-structured, and efficient approach                  | Mostly logical approach with minor gaps                                      | Basic approach with some inconsistencies          | Illogical or unclear approach                          |
| Accuracy of Solution             | 20% | Completely correct solution with proper justification             | Mostly correct with minor errors                                             | Partially correct solution                        | Incorrect or incomplete solution                       |
| Clarity                          | 10% | Clear, well-organized, and properly explained solution            | Understandable with minor clarity issues                                     | Limited clarity of explanation                    | Poorly presented and difficult to understand           |

### Marks Evaluation:

| Question No.               | Understanding of Problem Context (20%) | Application Concepts | Problem-Solving Approach (30%)<br>(20%) | Accuracy of Solution (20%) | Clarity (10%) | Total Marks |
|----------------------------|----------------------------------------|----------------------|-----------------------------------------|----------------------------|---------------|-------------|
| <b>Q1</b><br>(50)          | Marks)                                 |                      |                                         |                            |               |             |
| <b>Q2</b><br>(50)          | Marks)                                 |                      |                                         |                            |               |             |
| <b>Overall Total (100)</b> |                                        |                      |                                         |                            |               |             |

|                   |                    |                    |
|-------------------|--------------------|--------------------|
| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|

|                  |                  |                  |
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| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____      | Date: _____      | Date: _____      |